

LAB TASKS

Task # 01:

Write an Assembly program that stores three 8-bit values in registers. Use AND, OR, and XOR instructions to clear all odd bits, set all even bits, and then toggle all bits. Finally, use the TEST instruction to check whether bit 0 is set in the final result and display "Bit 0 is ON" or "Bit 0 is OFF" using conditional jumps.

Task # 02:

Write an Assembly program that takes three numbers from the user and finds the largest number two times — first using unsigned comparison (JA, JB, JAE) and then using signed comparison (JG, JL, JGE). Display both results with messages like "Largest (Unsigned) = ..." and "Largest (Signed) = ...".

Task # 03:

Write an Assembly program that reads up to 5 numbers from the user. If the user enters 0, the program should stop taking input immediately. Use ECX as a counter and control the loop using LOOPNZ or LOOPNE. After the loop ends, display how many valid (non-zero) numbers were entered.

Task # 04:

Write the following C program and then convert it into Assembly language using CMP and conditional jump instructions (JG, JL, JGE, etc.):

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    if (a > b && a < 100)
        printf("Condition True\n");
    else
        printf("Condition False\n");

    return 0;
}
```